CHAPTER L-08

LIFE INSURANCE UNDERWRITING

Chapter Introduction

A life insurance agent's work does not stop once a proposal is secured from a prospective customer. The proposal must also be accepted by the insurance company and result in a policy.

Every life insurance proposal has to pass through a gateway where the life insurer decides whether to accept the proposal and if so, on what terms. In this chapter we shall know more about the process of underwriting and the elements involved in the process.

Learning Outcomes

A. Underwriting – Basic concepts

B. Non-medical underwriting

C. Medical underwriting

A. Underwriting – Basic concepts

1. Underwriting purpose

Underwriting has two purposes

i. To assess the risk, classify the risk and decide the terms of acceptance or to decline the risk.
ii. To prevent anti-selection against the insurer

Definition

The term **underwriting** refers to the process of evaluating each proposal for life insurance in terms of the degree of risk it represents and then deciding whether or not to grant insurance and on what terms.

Anti-selection is the tendency of people, who suspect or know that their chance of experiencing a loss is high, to seek out insurance with a view to gain in the process.

Example

If life insurers were to be not selective about whom they offered insurance, there is a chance that people with serious ailments like heart problems or cancer, who did not expect to live long, would seek to buy insurance.

In other words, if an insurer did not exercise underwriting discretion, it would be selected against and may suffer losses in the process.

2. Equity among risks

The term "Equity" means that applicants who are exposed to similar degrees of risk must be placed in the same premium class. The Mortality table, used to determine premiums, represents the mortality experience of standard lives or average risks. They include the vast majority of individuals who propose to take life insurance.

a) Risk classification

To usher equity, the underwriter engages in a process known as **risk classification** i.e. individual lives are categorised and assigned to different risk classes depending on the degree of risks they pose. There are four such risk classes.

Diagram 1: Risk classification

i. Standard lives

These consist of those whose anticipated mortality corresponds to the standard lives represented by the mortality table.

ii. Preferred risks

These are the ones whose anticipated mortality is significantly lower than standard lives and hence could be charged a lower premium.

iii. Substandard lives

These are the ones whose anticipated mortality is higher than the average or standard lives, but are still considered to be insurable. They may be accepted for insurance with higher (or extra) premiums or subjected to certain restrictions.

iv. Declined lives

These are the ones whose impairments and anticipated extra mortality are so great that they could not be provided insurance coverage at an affordable cost. Sometimes an individual's proposal may also be temporarily declined if he or she has been exposed to a recent medical event, like an operation.

3. Underwriting process

Underwriting process takes place at two levels:

- ✓ At Field level
- ✓ At Underwriting department level

a) Field or Primary level

Field level underwriting is also known as **primary underwriting**. It includes information gathering by an agent or company representative to decide whether an applicant is suitable for granting insurance coverage. The agent plays a critical role as primary underwriter. He is in the best position to know the life to be insured.

Many insurance companies may require that agents complete a statement or a confidential report, asking for specific information, opinion and recommendations to be provided by the agent with respect to the proposed life.

Fraud monitoring and role of agent as primary underwriter

Much of the decision with regard to acceptance of a risk depends on the facts that have been disclosed by the proposer in the proposal form. It may be difficult for an underwriter who is sitting in the underwriting department to know whether these facts are untrue and have been fraudulently misrepresented with deliberate intent to deceive.

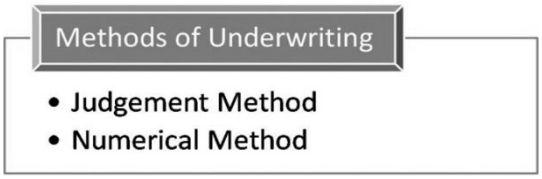
The agent plays a significant role here. He or she is in the best position to ensure that the facts that have been represented are true, due to his/ her direct and personal contact with the proposed life.

b) Underwriting at the Department level

The main level of Underwriting is at the Department or Office level. It involves specialists and persons who consider all the relevant data on the case to decide whether to accept a proposal for Life insurance and on what terms.

4. Methods of underwriting

Diagram 2: Methods of Underwriting



Underwriters may use two types of methods for the purpose:

Judgment Method	Numerical Method
Under this method subjective judgment is used, especially when deciding on a case that is complex.	Under this method underwriters assign positive rating points for all negative or adverse factors (negative points for any positive or favourable factors).
Example: Deciding whether life insurance can be given to a person staying in a disturbed country/ area.	**Example:** A person with history of cardiac ailments and/ or early deaths in the family may be assigned positive points. The total number of points so assigned will help an underwriter in deciding the extent of risk involved.
In such situations, the department may get the expert opinion of a medical doctor who is also called a medical referee.	The sum total of these positive/negative points, and/or is referred to as Extra Mortality Rating (EMR). Higher EMR indicates that the life is substandard. If the EMR is very high, underwriters may decline insurance.

Underwriting Decisions

Let us now consider the various kinds of decisions that underwriters may take with regard to a life proposed for underwriting.

a) **Acceptance at ordinary rates (OR)** is the most common decision. This rating indicates that the risk is accepted at the same rate of premium as would apply to an ordinary or standard life.

Diagram 3: Underwriting decisions

b) **Acceptance with an extra:** This is the most common way of dealing with the large majority of sub-standard risks. It involves charging an extra over the tabular rate of premium.

c) **Acceptance with a lien on the sum assured:** A lien is a kind of hold which the life insurance company can exercise (in part or whole) on the amount of benefit it has to pay in the event of a claim.

   Example: Consider the case of an insured who has suffered and recovered from a certain disease like TB. Imposition of Lien would imply that if this person were to die from a relapse of the TB, within a given period, only a decreased amount of death benefit may be payable.

d) **Acceptance with a restrictive clause:** For certain kinds of hazards a restrictive clause may be applied which limits death benefit in the event of death under certain circumstances.

   Example is a pregnancy clause imposed on pregnant ladies that limits insurance payable in the event of pregnancy related deaths occurring within say three months of delivery.

e) **Decline or postpone:** Finally, a life insurance underwriter may decide to decline or reject a proposal for insurance. This would happen when there are certain health/ other features which are so adverse that they considerably increase the risk.

   Example: An individual who suffers from cancer and has little chance of remission, would be a candidate for rejection,

Similarly in some cases it may be prudent to postpone acceptance of the risk until such time as the situation has improved and become more favourable.

Example

A lady who has just had a hysterectomy operation may be asked to wait for a few months before insurance on her life is allowed, to allow any post operation complications that may have arisen to disappear.

Test Yourself 1

Which of the following cases is likely to be declined or postponed by a life insurer?

I. A healthy 18 year old
II. A sports person
III. A person suffering from AIDS
IV. A housewife with no income of her own

B. Non-medical underwriting

1. Non-medical underwriting

A large number of life insurance proposals may typically get selected for insurance without conducting a medical examination to check the insurability of a life to be insured. Such cases are termed as **non-medical proposals**.

In view of multiple reasons including the costs involved, in some types of policies, Life insurers grant insurance without insisting on a medical examination

2. Conditions for non-medical underwriting

However non-medical underwriting calls for conditions like applicability to certain class of lives, certain plans of insurance, certain upper limits of sum insured, entry age limits, maximum term of insurance etc.to be followed.

3. Rating factors in underwriting

Rating factors refer to various aspects related to financial situation, life style, habits, family history, personal history of health and other personal circumstances in the prospective insured's life that may pose a hazard and increase the risk. Underwriting involves identifying these hazards and their likely impact and classifying the risk accordingly.

Rating factors may be broadly divided into two – those which contribute to moral hazard and those which contribute to physical [medical] hazards. Life insurance companies often divide their underwriting into categories accordingly. Factors like income, occupation, lifestyle and habits, which contribute to moral hazard, are assessed as part of **financial underwriting**, while medical aspects of health fall under **medical underwriting**.

a) Female insurance

Women generally have greater longevity than men. However they may face some problems with respect to moral hazard. This is because many women in Indian society are victims of male domination and social exploitation. Evils like dowry deaths exist even today. Longevity of women can also be affected from problems connected with pregnancy.

Insurability of women is governed by need for insurance and capacity to pay premiums. Insurance companies may thus decide to grant full insurance only to those who have earned income of their own and may impose limits on other categories of women. Similarly some conditions may be levied on pregnant women.

b) Minors

Minors have no contracting power of their own. Hence a proposal on the life of a minor has to be submitted by another person who is related to the minor in the capacity of a parent or legal guardian. It would also be necessary to ascertain the need for insurance, since minors usually have no earned income of their own. Three conditions would generally be sought when considering insurance for minors:

i. **Whether they have a properly developed physique**

   Poor physique can be a result of malnutrition or other health problems posing grave risks.

ii. **Proper family history and personal history**

   If there are adverse indicators here, it may pose risks.

iii. **Whether the family is adequately insured**

   It is necessary to check if the family has a culture of insurance. One must be on guard if no other member of the minor's family has been insured. Amount of insurance is generally linked to that of parents.

c) Large sums assured

An underwriter needs to be wary when the amount of insurance is very large relative to annual income of the proposed insured. Generally sum assured may be assumed to be around ten to twelve times one's annual income. If the ratio is much higher than this, it raises the possibility of selection against the insurer.

Example

If an individual has an annual income of Rs. 5 lakhs and proposes for a life insurance cover of Rs. 3 crores, it raises a cause for concern.

Typically concerns can arise in such instances because of the possibility that such a large amount of insurance is being proposed in anticipation of suicide or as a result of expected deterioration in health. A third reason for such large sums could be excessive misselling by the sales person.

Large sums assured would also mean premiums increasing in proportion and raise the question of whether the payment of such premiums would be continued. In general, the premium payable should be within one third of an individual's annual income

d) Age

Mortality risk is closely related to age. The underwriter needs to be careful when considering insurance for people of advanced ages.

Example

If the insurance is being proposed for the first time after age 50, there is a need to suspect moral hazard and enquire about why such insurance was not taken earlier.

We must also note that chances of occurrence of degenerative diseases like diseases of the heart and kidney failure increase with age and become higher at older ages. Life insurers may also seek for some special reports when proposals are submitted for high sums assured/ advanced ages or a combination of both.

Example

Examples of such reports are ECG; EEG; X-Ray of the chest and Blood Sugar test. These tests may reveal deeper insights about the health of the proposed life than the answers given in the proposal or an ordinary medical examination can provide.

Examples

When a proposal is submitted at a branch located far away from the place of residence of the proposed insured

A medical examination is done elsewhere even when a qualified medical examiner is available near one's place of residence.

A third case is when a proposal is made on the life of another without having clear insurable interest, or when the nominee is not the near dependent of the life proposed.

In each such case an enquiry may be made. Finally, when the agent is related to the life assured a moral hazard report may be called from a branch official like the agency manager/ development officer.

e) Occupation

Occupational hazards can arise from three sources:

- ✓ Accident
- ✓ Health hazard
- ✓ Moral hazard

Diagram 4: Sources of Occupational Hazards

i. Accidental hazards arise because certain kinds of jobs expose one to the risk of accident. There is any number of jobs in this category – like circus artistes, scaffolding workers, demolition experts and film stunt artistes.

ii. Health hazards arise when the nature of the job is such as to give rise to possibility of medical impairment. There are various kinds of health hazards.

- ✓ Some jobs like that of **rickshaw pullers** involve a lot of physical strain and impact the respiratory system.
- ✓ Situations where one may be exposed to **toxic substances** like mining dust or carcinogenic substances (that cause cancer) like chemicals and nuclear radiation.
- ✓ Working in **high pressure environments** like underground tunnels or deep sea, can cause acute decompression sickness.
- ✓ Finally, **overexposure** to certain job situations (like sitting cramped before a computer or working in a high noise setting) can impair functioning of certain body parts in the longer run.

iii. **Moral hazard** can arise when a job involves proximity or can cause predisposition towards criminal elements or to drugs and alcohol. An example is that of a dancer in a nightclub or an enforcer in a liquor bar or the 'bodyguard' of a businessman with suspected criminal links. Again the job profiles of certain individuals like superstar entertainers may lead them to intoxicating lifestyles, which sometimes come to tragic ends.

When an occupation falls under any such hazardous category, the applicant for insurance may need to complete an occupational questionnaire that asks for specific details of the job, duties involved and risks exposed to. A rating may also be imposed for occupation in the form of a flat extra (for example Rupees two per thousand sums assured.) Such extra may be reduced or removed when the insured's occupation changes.

f) Lifestyle and habits

Lifestyle and habits are terms, covering a wide range of individual lifestyle characteristics, which may be brought out in the agent's confidential reports and moral hazard reports, suggesting an exposure to risk. In particular three features are important:

Smoking and tobacco use: Use of tobacco is not only a risk in itself but also contributes to increasing other medical risks. Companies charge differential rates today for smokers and non-smokers and users of other forms of tobacco usage like *gutkha* and *paan masala*.

Alcohol: Drinking alcohol occasionally or in modest quantities is not considered a hazard. However, long term heavy drinking can impair liver functioning, affect the digestive system and lead to mental disorders. Alcoholism is also linked with accidents, violence, family abuse, depression and suicides.

Substance abuse: Substance abuse refers to the use of various kinds of substances like drugs or narcotics, sedatives and other similar stimulants. Some of these are even illegal and their use indicates criminal disposition and moral hazard.

Test Yourself 2

Which of the following is an example of moral hazard?

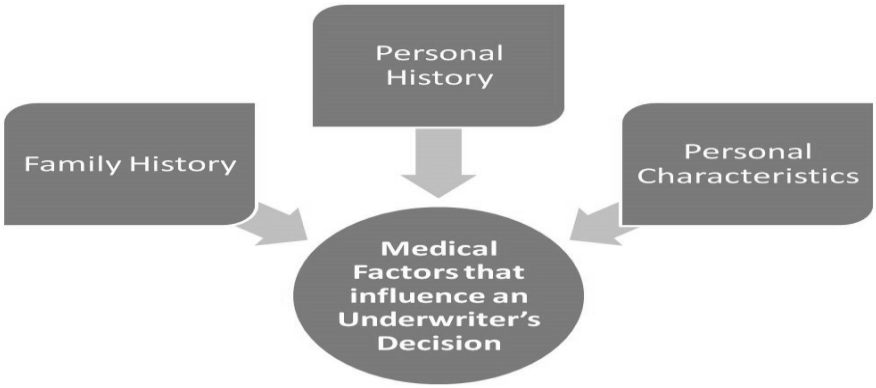
I. Stunt artist dies while performing a stunt
II. A person drinking copious amounts of alcohol because he is insured
III. Insured defaulting on premium payments
IV. Proposer misplacing policy document

C. Medical underwriting

1. Medical underwriting

Let us now consider some of the medical factors that would influence an underwriter's decision. These are generally assessed through medical underwriting. They may often call for a medical examiner's report. Let us look at some of the factors that are checked.

Diagram 5: Medical Factors that influence an Underwriter's Decision



a) Family history

The impact of family history on mortality risk has been studied from three angles.

i. **Heredity**: Certain diseases can be transmitted from one generation to another, say from parents to children.

ii. **Average longevity of the family**: When the parents have died early on account of certain diseases like heart trouble or cancer, it may be a pointer that the offspring may also not live long.

iii. **Family environment**: Thirdly, the environment in which the family lives can cause exposure to infection and other risks.

Life insurers have thus to be careful when entertaining cases of individuals with adverse family history. They may call for other reports and may impose an extra mortality rating in such cases.

b) Personal history

Personal history refers to past impairments of various systems of the human body which the life to be insured has suffered from. The proposal form for life insurance typically contains a set of questions which enquire whether the life to be insured has been under treatment for any of these.

The major kinds of ailments that are considered by the underwriters include Cardiovascular diseases, diseases of the respiratory system, malignant tumours/ cancer, ailments of the renal system, impairments of the endocrine system, diseases of the digestive system like gastric ulcers and cirrhosis of the liver and diseases of the nervous system.

c) Personal characteristics

These can also be significant indicators of the tendency to disease.

i. Build

A person's build consists of his height, weight, chest and girth of the abdomen. For given age and height, there is a standard weight that has been defined and if the weight is too high or low in relation to this standard weight, we can say that the person is overweight or underweight.

Similarly, it is expected that the chest should be expanded at least by four centimetres in a normal person and that the abdominal girth should not be more than one's expanded chest.

ii. Blood pressure

Another indicator is a person's blood pressure. There are two measures of this

- ✓ Systolic
- ✓ Diastolic

When the actual readings are much higher than the normal values, we say that the person has high blood pressure or hypertension. When it is too low, it is termed as hypotension. The former can have serious consequences.

iii. Urine – Specific gravity

Finally, a reading of the specific gravity of one's urine can indicate the balance among various salts in the urinary system. It can indicate any malfunctioning of the system.

d) Backdating:

Backdating means changing the start date of the policy to an earlier one. For example, you bought a Life insurance policy on 1st June, 2013 but later you think that the policy would have generated better returns if you had bought it in April 2013. You and your insurance company agree to change the policy to officially start it from April, 2013. In this case, you have backdated the policy. Usually, no interest is charged if the policy is backdated by less than a month.

Backdating is done for the following purposes:

(i) **Getting a lower premium based on age:** While issuing the policy, insurers consider the nearest age of the policyholder. It means if you are 32 years and 7 months old, the insurer will consider your age as 33 years. This nearest age may put you in a higher premium slab. However, if you backdate the policy by 2 months, the insurer will consider your age as 32 years and 5 months only. Now you will be paying lower premiums based on a plan for a 32-year old.

(ii) **Set the timing of payment:** There are specific professions where the income flow is not steady. In such a scenario if an individual accidently buys a life insurance policy in its off-season then the policy can be backdated to the period of maximum earnings. For instance, a farmer may have a seasonal income. He would prefer to make insurance payments only after he has received his crop proceedings. In this case, a farmer could backdate the policy to start it in the harvest season.

(iii) **To coincide with special dates:** You can backdate the policy to coincide with your important dates, such as birthday and anniversary. It keeps easy for you to remember your premium due date.

(iv) **Early maturity claims**: Backdating reduces the tenure of a policy and facilitates early maturity. For instance, if a 30-year life insurance cover bought on March 2000 is backdated to April 1999, the policy would mature on April, 2029 instead of March 2030. In case of endowment policies, this could be beneficial as maturity benefits accrue earlier.

Test Yourself 3

Why is heredity history of importance in medical underwriting?

I. Rich parents have healthy kids
II. Certain diseases can be passed on from parents to children
III. Poor parents have malnourished kids
IV. Family environment is a critical factor

Summary

- To bring equity, the underwriter engages in risk classification where individual lives are categorised and assigned to different risk classes depending on the degree of risks they pose.
- Underwriting process may be said to take place at two levels:
  - ✓ At field level and
  - ✓ At underwriting department level
- Underwriting decisions made by underwriters include acceptance of standard risk at standard rates or charging extra for sub-standard risks. Sometimes there is acceptance with lien on sum assured or acceptance is based on restrictive clauses. Where the risk is large the proposal is declined or postponed.

- A large number of life insurance proposals may typically get selected for insurance without conducting a medical examination. Such cases are termed as non-medical proposals.
- Some of the rating factors for non-medical underwriting include
  - ✓ Age
  - ✓ Large sum assured
  - ✓ Moral hazard etc.
- Some of the factors considered in medical underwriting include
  - ✓ Family history,
  - ✓ Heredity and personal history etc.

Key Terms

1. Underwriting
2. Standard life
3. Non-medical underwriting
4. Rating factor
5. Medical underwriting
6. Anti-selection

Answers to Test Yourself

Answer 1 - The correct option is III.

Answer 2 - The correct option is II.

Answer 3 - The correct option is II.